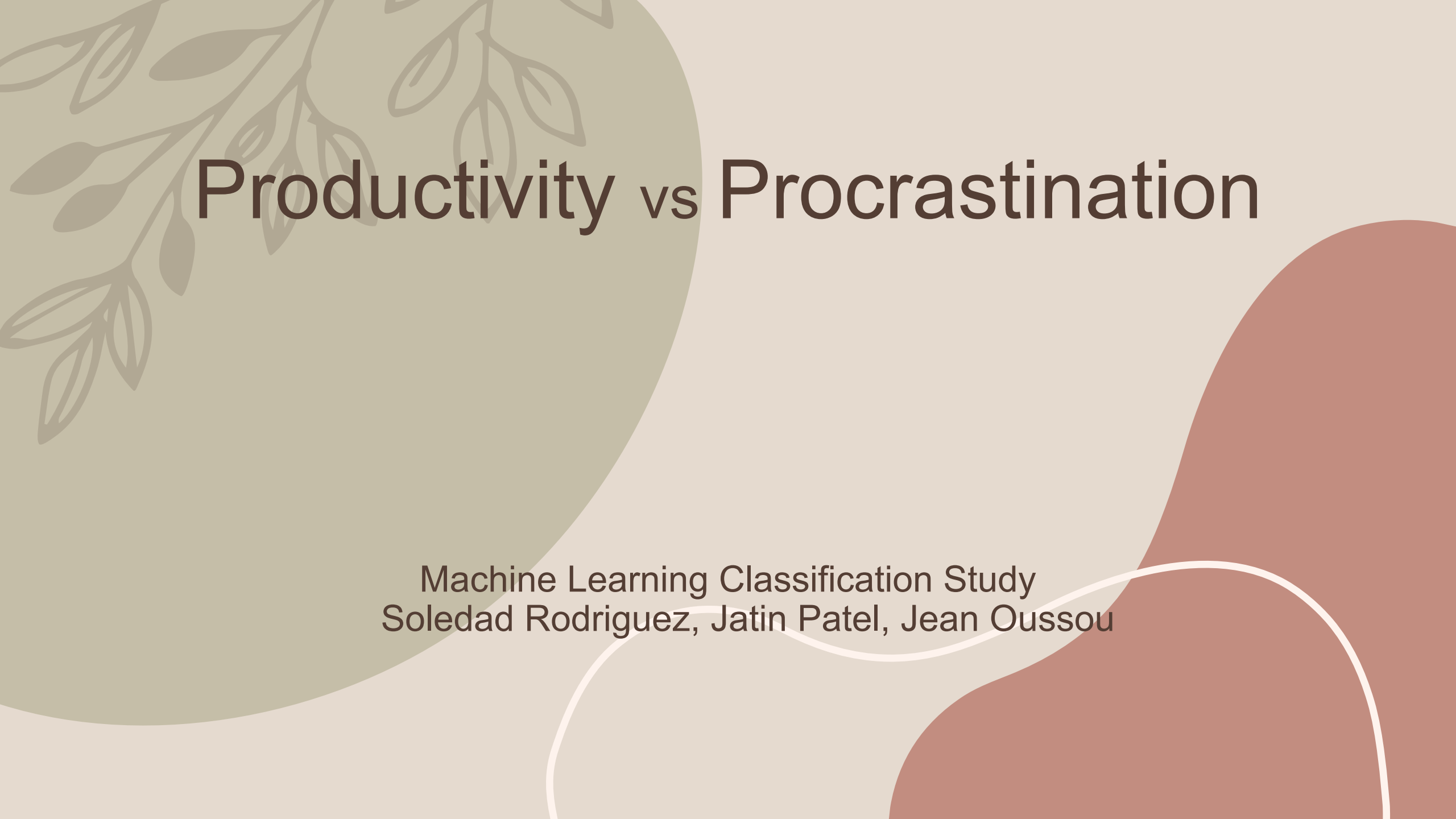




Productivity vs Procrastination

Machine Learning Classification Study
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PROBLEM

- In today's fast-paced world, productivity plays a key role in academic and professional success.
- However, many individuals struggle with procrastination, which leads to delays, lower performance, and increased stress.
- People often assume that productivity depends only on discipline or motivation—but is that really true?

PROBLEM (CONT.)

To determine whether productivity can be accurately predicted, we analyze a dataset that captures behavioral patterns such as study habits, time management, and daily activities.

Using machine learning techniques, we aim to understand how these factors influence productivity levels and whether accurate predictions can be made.

DATASET OVERVIEW

The dataset used in this project comes from Kaggle and includes features related to study habits, time management, and behavior.

It was prepared and analyzed to understand how different factors influence productivity levels.

Data Description

3 TOTAL FEATURES

- Behavioral attributes
- Study habits
- Time management data

INSTANCES

- 400 samples

TARGET

- Productivity level
 - High
 - Medium
 - Low

DATA TYPE

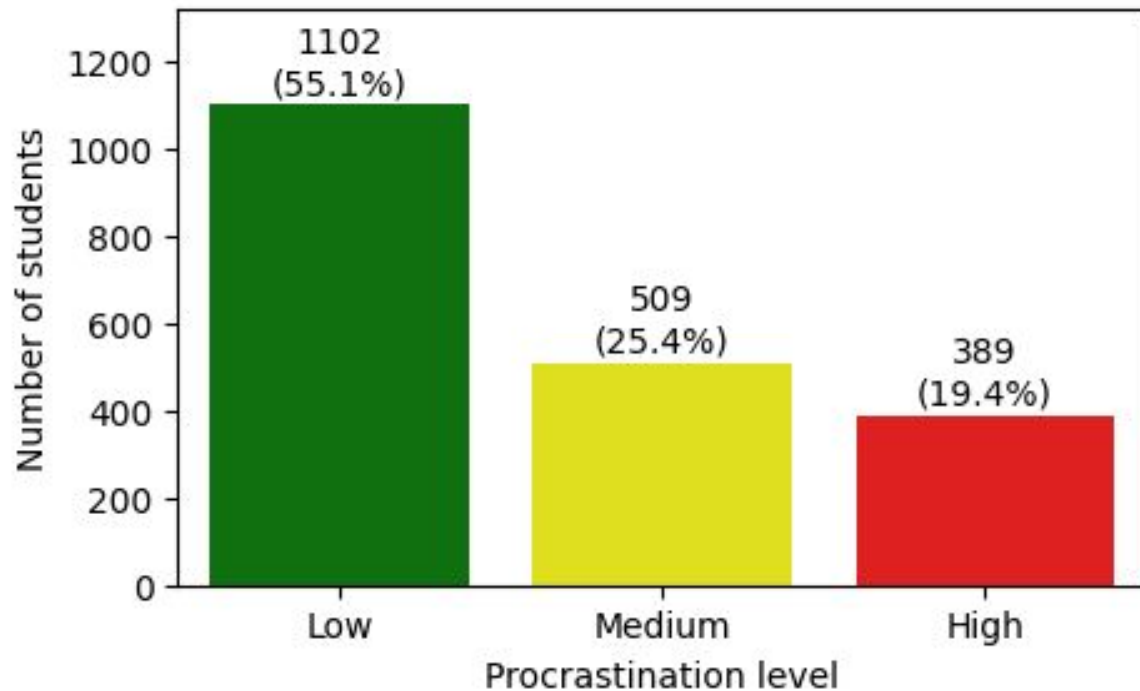
- Numerical and categorical features

DA FINDINGS

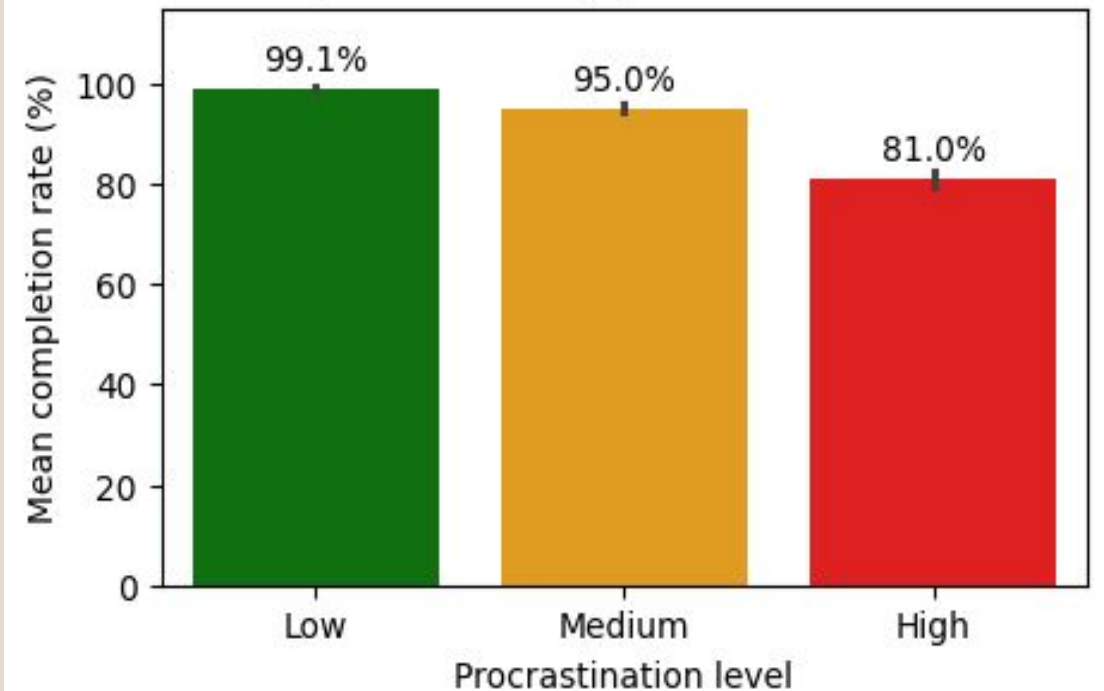
- Dataset contains both numerical and categorical features
- Medium productivity has the highest number of samples
- Some overlap exists between Medium and High productivity
- Behavioral patterns strongly influence productivity levels

VISUALIZATION

Procrastination levels



Completion rate by procrastination level



DATA PREPROCESSING

- Removed irrelevant features
- Handled missing values
- Converted categorical variables into numerical form
- Split data into training and testing sets

CLASSIFIERS

- o **Random Forest Classifier**

Random Forest is a machine learning classifier that makes predictions by combining multiple decision trees.

- o **Decision Tree**

Decision Tree is a model that makes predictions by splitting the data based on feature values.

- o **Standard Logistic Regression**

Logistic Regression is a classification algorithm that predicts the probability of different classes.

Results

- Random Forest achieved the highest accuracy of **82%**, making it the best-performing model
- Logistic Regression showed stable and consistent performance across all classes
- Decision Tree performed reasonably well but had lower accuracy compared to other models
- Overall, the models were effective in predicting productivity levels

Model	Accuracy
Random Forest	82%
Decision Tree	~75%
Logistic Regression	~78%

COMPARISON

- Random Forest outperformed the other models in terms of accuracy and overall performance
- Logistic Regression showed consistent results but was slightly less accurate than Random Forest
- Decision Tree had lower accuracy and was more sensitive to variations in the data
- Overall, ensemble methods like Random Forest provide better performance compared to single models

Model	Performance	Accuracy
Random Forest	Best	82%
Logistic Regression	Good	~78%
Decision Tree	Moderate	~75%

CONCLUSIONS

- Machine learning models can effectively predict productivity levels based on behavioral patterns
- Features such as study habits and time management play a significant role in determining productivity
- Random Forest achieved the best performance due to its ability to combine multiple decision trees
- The results show that data-driven approaches can provide useful insights into productivity analysis